Shashank H R

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SUMMARY

Back-end Software Engineer with **2+ years of experience** building, load-testing, and scaling Java microservices and distributed systems (**Spring Boot**) on a high-volume B2B messaging platform. Strong in performance and scalability: k6 load testing, fault injection, connection-pool and cache tuning, and Kafka capacity planning. Owns features end-to-end, from design to production troubleshooting.

TECHNICAL SKILLS

Languages: Java, Python, SQL, Data Structures and Algorithms
Frameworks: Spring Boot, Quarkus, REST APIs, Microservices Architecture, JUnit, Design Patterns (Factory, Strategy)
Core Concepts: Distributed Systems, System Design, Multithreading and Concurrency (Java Virtual Threads)
Messaging and Data: Apache Kafka, Kafka Connect, RabbitMQ, Redis, MongoDB, MySQL, Elasticsearch
Performance: k6 Load Testing, Fault Injection, Capacity Planning, OpenTelemetry, Grafana
Infrastructure: Docker, Nginx, Jenkins, Gradle, Git, CI/CD, AWS (EC2, S3, OpenSearch), Agile (Scrum)

PROFESSIONAL EXPERIENCE

Software Engineer Jul 2024 - Present
Engati *Bangalore, India*

- Led performance validation of a microservices-based WhatsApp Business API platform (Quarkus, Java 21, REST APIs): built a k6 harness with a configurable Meta mock and ran **30+ documented test cases** (baseline, slow-tail, timeout, connection-reset, 429, 5xx incident, ramp, ceiling, spike-recovery), keeping **p95 under 1 s up to 742 virtual users**, monitored with Grafana and OpenTelemetry.
- Root-caused a hidden connection cap (the REST client ignored the configured pool size) and tuned the API tier for scalability with virtual threads, identity caching, and lower-volume logging; established a ~**713 RPS single-pod ceiling** and a **500 RPS planning target** with 30% headroom, documenting verified pool and timeout settings.
- Reconciled **1.1M webhook packets** across the distributed delivery pipeline against attempts, retries, and terminal outcomes, proving no unexplained loss (retry amplification 1.052 vs 1.0525 expected).
- Built a Kafka partition-sizing model for the webhook service (throughput, pod count, burst) for a distributed delivery service; showed 4 partitions suffice through 2027, set a 4-6-12-24 growth ladder, and a pods-vs-partitions guard.
- Split webhook ingestion from customer delivery into two REST and Kafka-based microservices with independent topics and consumer groups and a staged cutover plan; added credit-exhaustion gating (200 ms Redis lookup, fail-open) that stops outbound sends without dropping conversations.
- Built the RCS billing pipeline: webhooks to Kafka, into AWS S3 via a **Kafka Connect** S3 sink, aggregated per customer and metric by idempotent Spark jobs; load-tested the connector at **up to 4M billing events/day**.
- Designed a fault-tolerant auto-retry framework in Spring Boot (Factory and Strategy patterns for pluggable back-off) for third-party REST API integrations (LeadSquared, MoEngage) using Redis, MongoDB, and RabbitMQ, handling ~2M API triggers/day; **recovers 45% of failed deliveries on the first retry**.
- Engineered a Spring Boot AI code review service (GitLab, Slack, multi-LLM, SQL, Docker, Jenkins CI/CD) with parallel virtual-thread chunk review of diffs up to 14K lines; adopted by ~20 developers and **cut review time from ~2 hours to 30–60 minutes a day per developer**; recognised with a **company award**.
- Root-caused repeated memory spikes in a Kafka-consumer Spring Boot service using heap and thread dumps: a misconfigured `@Async` thread pool (sizing and queue limits) caused thread churn and DB `SQLException` failures; right-sized it, then applied the same dump analysis to **optimise memory in 2–3 production Java applications**.
- Cut MongoDB (M20) memory use **over 60%** with missing indexes (scan-to-return 3,000:5 to under 100:1); fixed a FastAPI (Python) leak, **peak memory 2.5 GB to ~1 GB**.

Software Engineer Intern Jan 2024 - Jun 2024
Engati *Bangalore, India*

- Built Spring Boot e-commerce lead-capture and abandoned-cart workflows, using the Factory pattern to extend them to checkout and CTA notifications.

EDUCATION

Bachelor of Engineering (B.E.), Information Science 2020 - 2024
JSS Science and Technology University, Mysuru CGPA: 9.47/10.0
Coursework: Data Structures and Algorithms, DBMS, Operating Systems

CERTIFICATIONS

MongoDB Associate Developer, MongoDB Inc. credly.com/badges/9dcb0453-4c7e-43b2-94e9-33017a70c617